

UNIT - IV WAVE MECHANICS

Syllabus

Introduction to matter waves de-Broglie's wavelength, Uncertainty principle, Schrodinger's wave equation-time dependent and time independent, Application to particle in a box, Barrier penetration problem, Tunneling effect and tunnel diode.

INTRODUCTION

4.1. DE-BROGLIE CONCEPT OF MATTER WAVES

Matter and radiation are the fundamental entities of the nature. According to De-Broglie, the matter, like radiation has dual nature, i.e., particle-like and wave like characteristics. All the moving matter particles like electron, proton, neutron, atom or molecules have an associated wave with them called as matter wave or **De-Broglie wave**. It is observed in experiments related to interference, diffraction, and polarization that light shows wave behavior and in experiments such as photoelectric effect and Compton Effect, it shows particle behavior. This dual behavior is known as wave –particle duality of light.

Thus, a light radiation has a dual nature, i.e., it possesses both particle wave properties, but it does not appear to behave as a wave and a particle simultaneously.

Thus a moving particle is always associated with wave. The particle is controlled by these waves and they are called **matter waves**.

4.2. DE-BROGLIE JUSTIFIED HIS CONCEPT WITH THE FOLLOWING POINTS:

- Our universe is fully composed of light and matter.
- Nature prefers symmetry. If radiation like light can act like wave sometime and like a particle at other time, then material particles (e.g., Electron, Neutron etc.,) should also act like wave at some time and particle in some other times.
- Every moving particle has always associated with wave. The particle is controlled by this wave.

4.3. De Broglie's Wavelength of Matter Waves:

De Broglie equated the energy equations of Planck law (wave) and Einstein mass-energy relation (particle). For a wave of frequency 'v', the energy associated with each photon is given by **Planck's relation**,

$$E = h\nu \dots\dots\dots (1)$$

Where, h is Planck's constant.

According to **Einstein's mass energy relation**, a mass 'm' is equivalent to energy,

$$E = mc^2 \dots\dots\dots (2)$$

Where, 'c' is the velocity of light.

From eqn. (1) and (2) $h\nu = mc^2$

(Since $\nu = c/\lambda$), Therefore $\frac{hc}{\lambda} = mc^2$

$$\lambda = \frac{h}{mc} \dots\dots\dots (3)$$

For a particle moving with a velocity 'v', if $c = v$

From equation (3) $\lambda = \frac{h}{mc} = \frac{h}{p}$

Where $p = mv$, the momentum of the particle.

This is **de-Broglie** wave equation, can be applied to a material particle and photons.

4.3.1. DE-BROGLIE WAVE LENGTH IN TERMS OF ITS ENERGY

If 'E' is the kinetic energy of the material particle then

$$E = \frac{1}{2} mv^2 \quad \dots\dots\dots (1)$$

(Multiply and divide by m in RHS)

$$= \frac{m^2 v^2}{2m} \quad \dots\dots\dots (2)$$

$$m^2 v^2 = 2mE$$

$$p^2 = 2mE$$

$$p = \sqrt{2mE} \quad \dots\dots\dots (3)$$

Therefore De –Broglie wave length is $\lambda = \frac{h}{\sqrt{2mE}} \quad \dots\dots\dots (4)$

4.3.2. DE-BROGLIE WAVE LENGTH ASSOCIATED WITH ACCELERATED CHARGED PARTICLES (ELECTRONS)

If a charged particles, say an electron is accelerated by a potential difference of 'V' volts, then its kinetic energy is given energy by

$$K.E = e V$$

$$e V = \frac{1}{2} mv^2$$

$$2eV = mv^2$$

Multiply 'm' on both sides

$$2meV = m^2 v^2$$

Take sq. root on both sides

$$mv = \sqrt{2meV} \quad \dots\dots\dots (1)$$

Then, the wavelength of electron is given by

$$\lambda = h/m v \quad \dots\dots\dots (2)$$

Substitute eqn. (1) in (2), $\lambda = \frac{h}{\sqrt{2meV}}$

4.4. PROPERTIES OF MATTER WAVES

- Lighter is the particle, greater is the wavelength associated with it.
- Smaller is the velocity of the particle, greater is the wavelength associated with it.
- If $v = 0$, then $\lambda = \text{infinity}$, i.e., wave become indeterminate and if $v = \text{infinity}$ then $\lambda = 0$.

This indicates that the De-Broglie waves are generated by the motion of particle.

d) These waves are produced in the particles whether the particles are charged particles or uncharged particles (in the equation $\lambda = h/m v$, there is no term representing the charge. It is independent of charge). This shows that these waves are not EM waves but are a new kind of waves.

e) The velocity of matter wave is not constant since it depends on the

velocity of the material particle whereas the velocity of EM wave is constant.

f) The velocity of these waves is greater than velocity of light. They can travel faster than light.

4.5. HEISENBERG'S UNCERTAINTY PRINCIPLE

STATEMENT

The Heisenberg uncertainty principle states that in any simultaneous determination of a pair of physical quantities which describes the motion of an atomic particle the product of uncertainties is equal to $h/2\pi$, i.e.,

$$\Delta x \cdot \Delta p_x \geq \frac{h}{2\pi} \quad \dots\dots\dots (1)$$

Where, Δx and Δp_x are the uncertainties in the determination of position and momentum respectively.

As a result, it is impossible to design an experiment to prove the wave and particle nature of matter at the same time. If one measures the position accurately, then the measurement of momentum becomes incorrect and vice versa. This principle gives only the probability of finding the particle in a given space in place of certainty.

$$\Delta E \cdot \Delta t \geq \frac{h}{2} \quad \dots\dots\dots (2)$$

Where, ΔE and Δt are the error in the measurement of energy and time respectively.

4.5.1. Experimental illustrations of uncertainty principle and Determination of the position of a particle by microscope

Illustration

Suppose we have to measure the position and the momentum of an electron by the use of an imaginary microscope having high resolving power. The electron can be resolved by the microscope only when a photon is scattered by an electron into the microscope lens.

The resolving power of the microscope is

$$\Delta x = \frac{\lambda}{2 \sin \theta} \quad \dots\dots\dots (1)$$

Where, Δx is the distance between any two points. It is the minimum position which can be resolved by the microscope when the photon is scattered by the electron. Therefore, Δx is taken as the uncertainty in the measurement of the position of the electron.

Let us consider the scattering of electron by a photon as shown in Fig 4.1. The electron is viewed by the microscope through AB. The momentum imparted by the photon on the electron is h/λ . The x-component of the momentum along OA is $(-h/\lambda) \sin \theta$. The x-component of the momentum along OB is $(h/\lambda) \sin \theta$

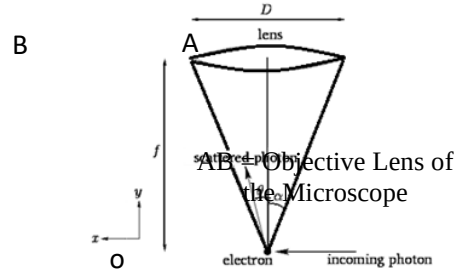


Fig:4.1.Scattering of the Electron by Photon

Therefore the, the change in momentum of the electron

$$\Delta p = \frac{h}{\lambda} \sin \theta - \left(-\frac{h}{\lambda} \sin \theta \right)$$

$$\Delta p = \frac{2h}{\lambda} \sin \theta \quad \dots\dots\dots (2)$$

The product of the uncertainties in position and momentum is

$$\Delta x \cdot \Delta p = \frac{2h}{\lambda} \sin \theta \times \frac{\lambda}{2 \sin \theta}$$

Therefore, $\Delta x \cdot \Delta p \approx h$

The product of the uncertainties treatment gives the correct values of the uncertainties of the position and momentum.

4.6. Physical significance of the wave function

The displacement 'y' determines the vibrations of a stretched string and 'E' determines the electric field intensities at different points. Similarly 'Ψ' measures the variations of the matter wave. Thus it connects the particle and its associated wave statistically.

1. The wave function or complex displacement 'Ψ' is a complex quantity and we cannot measure it.
2. The matter wave can be represented by wave function. This wave function is used to identify the state of a particle in an atomic structure.
3. It can tell the probability of the position of the particle at a time but it cannot predict the exact location of the particle at that time. Thus it tells us where the particle is likely to be not where it is.
4. We can say that the wave function as probability amplitude since it is used to find the location of the particle.
5. The probability of finding a particle in a particular volume element $d\tau$ is given by

$$P(r) d\tau = |\Psi^* \Psi| d\tau = |\Psi(r, t)|^2 d\tau$$

Where 'Ψ*' is called the complex conjugate of 'Ψ'.

$|\Psi|^2$ is proportional to probability density of the particle in the state Ψ . A large value of it indicates the strong probability of the body's presence while a small value of it means the slight probability of its presence.

6. $\iiint |\Psi|^2 d\tau = 1$ when the particle presence is certain in the space.
7. Being a complex function, it does not have a direct physical meaning, but when we multiply this with its complex conjugate, the product $|\Psi|^2$ has physical meaning. i.e., we will speak normally the intensity of light at a point rather than the amplitude of light at a point since intensity (square of the amplitude) is a measurable and real quantity. The same thing is true for matter waves also.

4.7. SCHRODINGER'S WAVE EQUATIONS

Schrodinger equations are the basic equations in quantum mechanics like Newton's law of motion. Schrodinger equation is used for both microscopic and macroscopic particles. Schrodinger describes the wave nature of a particle in mathematical form and is known as **Schrödinger wave equation**. In that we have two types, they are

- a) Time Dependent wave Equation
- b) Time Independent wave Equation

4.7.1. Time Independent wave Equation

Let us consider a particle of mass m , moving with a velocity ' v '. let ' Ψ ' be the wave function of the particle along x , y and z coordinates axis. The De-Broglie wavelength associated is given by

$$\lambda = \frac{h}{mv}$$

The classical differential equation of progressive wave moving with a wave velocity ' v ' can be written as

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots (1)$$

where

$$\begin{aligned} \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} &= \nabla^2 \text{ (Laplacian operator)} \\ \therefore \nabla^2 \Psi &= \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2} \dots\dots\dots (2) \end{aligned}$$

The solution for equation (2) is assumed to be

$$\Psi = \Psi_0 e^{-i\omega t} \dots\dots\dots (3)$$

Where ' Ψ_0 ' is the amplitude of the wave at the point (x, y, z)

Diff eqn. (3) w.r.t to ' t ' twice we get

$$\frac{\partial \Psi}{\partial t} = \Psi_0 e^{-i\omega t} \cdot (-i\omega)$$

$$\frac{\partial^2 \Psi}{\partial^2 t} = \Psi_0 e^{-i\omega t} (-i\omega)(-i\omega)$$

$$\frac{\partial^2 \Psi}{\partial^2 t} = i^2 \omega^2 \Psi_0 e^{-i\omega t}$$

Since, $i^2 = -1$ and $\Psi = \Psi e^{-i\omega t}$

$$\frac{\partial^2 \Psi}{\partial^2 t} = -\omega^2 \Psi \quad \dots\dots\dots (4)$$

Substituting (4) in (2) we get

$$\therefore \nabla^2 \Psi = -\frac{\omega^2 \Psi}{v^2}$$

$$\nabla^2 \Psi + \frac{\omega^2}{v^2} \Psi = 0 \quad \dots\dots\dots (5)$$

But, $\omega = 2\pi \nu = 2\pi \frac{v}{\lambda}$ [where $\nu = \frac{v}{\lambda}$]

$$\text{or } \frac{\omega}{v} = \frac{2\pi}{\lambda} \quad \dots\dots\dots (6)$$

Squaring eqn. (6) on both sides, we get

$$\text{or } \frac{\omega^2}{v^2} = \frac{4\pi^2}{\lambda^2} \quad \dots\dots\dots (7)$$

Substituting (7) in (5), we get

$$\nabla^2 \Psi + \frac{4\pi^2}{\lambda^2} \Psi = 0 \quad \dots\dots\dots (8)$$

By De-Broglie's equation we know $\lambda = \frac{h}{mv}$

$$\therefore \nabla^2 \Psi + \frac{4\pi^2 m^2 v^2}{h^2} \Psi = 0 \quad \dots\dots\dots (9)$$

If 'E' is the total energy of the particle, 'V' is the potential energy and $\frac{1}{2}mv^2$ the kinetic energy, then

$$\text{Total energy } E = \text{P.E} + \text{K.E}$$

$$E = V + \frac{1}{2}mv^2$$

$$E - V = \frac{1}{2}mv^2$$

$$2(E - V) = mv^2$$

Multiplying by 'm' on both sides in above equation we get,

$$2m(E - V) = m^2 v^2 \quad \dots\dots\dots (10)$$

Substituting (10) in (9) we get

$$\nabla^2 \Psi + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0 \quad \dots\dots\dots (11)$$

The above equation is known as **Schrödinger Time Independent wave equation**

For a one-dimensional motion, the Schrödinger equation takes the form

$$\frac{\partial^2 \Psi}{\partial x^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$$

Introducing $\hbar = \frac{h}{2\pi}$ we get

$$\nabla^2 \Psi + \frac{2m}{\hbar^2} (E - V) \Psi = 0 \quad \dots\dots\dots (12)$$

For free particle $V=0$, therefore Schrödinger wave equation becomes.

$$\nabla^2 \Psi + \frac{2mE}{\hbar^2} \Psi = 0 \quad \dots\dots\dots (13)$$

4.7.2. Time Dependent wave Equation

We can easily derive Schrodinger time dependent wave equation by eliminating E from Schrodinger time independent wave equation.

Differentiating the Eq. (14) with respect to time t, we get.

$$\Psi(x,y,z) = \Psi_o(x,y,z) e^{-i\omega t} \quad \dots\dots\dots (14)$$

$$\frac{\partial \Psi}{\partial t} = -i \omega \Psi_o e^{-i\omega t} \quad \dots\dots\dots (15)$$

$$= -i (2\pi\nu) \Psi_o e^{-i\omega t} \quad (\because \omega = 2\pi\nu)$$

$$\frac{\partial \Psi}{\partial t} = -i (2\pi\nu) \Psi_o e^{-i\omega t} \quad \dots\dots\dots (16)$$

$$= -i (2\pi\nu) \Psi$$

$$= -2\pi i \frac{E\Psi}{h}$$

$$(\because E = h\nu \text{ then } \frac{E}{h} = \nu) ; \left(\frac{h}{2\pi} = \hbar \right)$$

$$\therefore \frac{\partial \Psi}{\partial t} = -i \frac{E\Psi}{\hbar} \quad \dots\dots\dots (17)$$

Multiplying i on the both sides,

$$i \frac{\partial \Psi}{\partial t} = -i \times i \frac{E\Psi}{\hbar}$$

$$(\because i \times i = -1)$$

$$i \frac{\partial \Psi}{\partial t} = \frac{E\Psi}{\hbar}$$

$$\text{or} \quad E \Psi = i\hbar \frac{\partial \Psi}{\partial t} \quad \dots\dots\dots (18)$$

Substitute, The value of $E \Psi$ from Eq. (18) in Schrodinger time independent wave Eq. (14) we have

$$\begin{aligned} \nabla^2 \Psi + \frac{2m(E-V)\Psi}{\hbar^2} &= 0 \\ \nabla^2 \Psi + \frac{2m(E\Psi - V\Psi)}{\hbar^2} &= 0 \\ \nabla^2 \Psi + \frac{2m(i\hbar \frac{\partial \Psi}{\partial t} - V\Psi)}{\hbar^2} &= 0 \\ \nabla^2 \Psi &= - \frac{2m(i\hbar \frac{\partial \Psi}{\partial t} - V\Psi)}{\hbar^2} \\ \frac{-\hbar^2}{2m} \nabla^2 \Psi &= i\hbar \frac{\partial \Psi}{\partial t} - V\Psi \\ \frac{-\hbar^2}{2m} \nabla^2 \Psi + V\Psi &= i\hbar \frac{\partial \Psi}{\partial t} \quad \dots\dots\dots (19) \end{aligned}$$

The equation is known as **Schrodinger time dependent** wave equation. Eq. (19) may be written as

$$\begin{aligned} \left(\frac{-\hbar^2}{2m} \nabla^2 + V \right) \Psi &= i\hbar \frac{\partial \Psi}{\partial t} \\ H \Psi &= E \Psi \quad \dots\dots\dots (20) \end{aligned}$$

Where, $H = \left(\frac{-\hbar^2}{2m} \nabla^2 + V \right)$ is Hamiltonian operator and $E = i\hbar \frac{\partial}{\partial t}$ is the energy operator

4.8. Application of Schrödinger's equation to one dimensional box : (or) particle in a one-dimensional box.

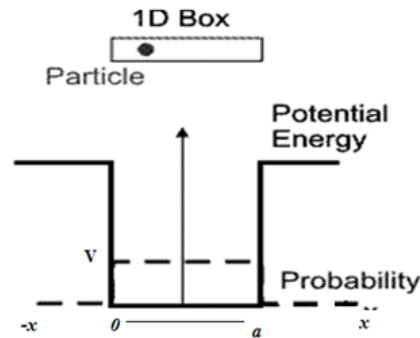


Fig:4.2. Particle in One Dimensional box

Let us consider a particle (e^-) of mass ' m ' moving along ' x -axis' enclosed in '1-D' potential box, since the walls are of potential the particle does not penetrate out the box. The particle is confined between the length ' a ' of the box and has elastic collisions with the walls. Therefore the potential energy of the electron inside the box is constant and can be taken as zero. We can say that outside the box and on the wall of the box, the potential energy ' V ' of the electron is infinity. Inside the box the potential energy ' (V) ' of the electron is zero.

To find the wave functions of the particle within the box of length ' l ' we will take the 1-D'

$$\text{Schrodinger time independent wave equation (i.e) } \frac{d^2 \Psi}{dx^2} + \frac{2m}{\hbar^2} (E-V) \Psi = 0 \quad \dots\dots\dots (1)$$

Since the potential inside the box is zero [(i.e) $V=0$] the particle has kinetic energy alone and thus it is named as a free particle (or) free electron.

∴ For a free particle, the Schrodinger wave Eq. is given by

$$\frac{d^2\Psi}{dx^2} + \frac{2m}{\hbar^2} E \Psi = 0 \quad \dots\dots\dots (2)$$

$$\frac{d^2\Psi}{dx^2} + k^2\Psi = 0 \quad \dots\dots\dots (3)$$

$$k^2 = \frac{2m}{\hbar^2} E \quad \dots\dots\dots (4)$$

Eqn. (3) is a second order differential equation. Therefore it should have solution with two arbitrary constants. The solution of Eqn. (3) is given by

$$\Psi(x) = A \sin kx + B \cos kx \quad \dots\dots\dots (5)$$

A, B are arbitrary constants, these constants can be found out by applying boundary conditions.

(i.e) $V(x) = \infty$. When 'x = 0' and 'x = 'a'

Applying Boundary conditions:

(i) At $x = 0$; Potential Energy $V = \infty$; there is no chance for finding the particle at the walls of the box.

$$\Psi(x) = 0$$

Applying these conditions in equation (5) we get

$$0 = A \sin 0 + B \cos 0$$

$$0 = 0 + B \quad (1)$$

$$B = 0 \quad \dots\dots\dots (6)$$

(ii) At $x = a$, P.E $V = \infty$;

∴ There is no chance for finding the particle at the walls of the box.

$$\Psi(x) = 0$$

Eq.(5) becomes

$$0 = A \sin ka + B \cos ka$$

From (6); $B = 0$

$$0 = A \sin ka$$

$$\sin ka = 0 ; \text{ since } A \neq 0$$

$$ka = n\pi$$

$$k = \frac{n\pi}{a}, \text{ where, } n=0,1,2,3 \quad \dots\dots\dots (7)$$

Substituting Eqn. (6) and Eqn. (7) in Eqn. (5) we get

$$\Psi(x) = A \sin \frac{n\pi x}{a} \quad \dots\dots\dots (8)$$

Energy of the Particle

From Eqn. (4)
$$k^2 = \frac{2mE}{\hbar^2} = \frac{2mE}{\frac{h^2}{4\pi^2}}$$

$$k^2 = \frac{8\pi^2 E m}{h^2} \dots\dots\dots (9)$$

Squaring Eqn. (7) we get

$$k^2 = \frac{n^2 \pi^2}{a^2} \dots\dots\dots (10)$$

Equating the Eqn. (9) and (10) we get

$$\frac{8\pi^2 E m}{h^2} = \frac{n^2 \pi^2}{a^2}$$

Energy of the particle
$$E_n = \frac{n^2 h^2}{8ma^2} \dots\dots\dots (11)$$

From Eqn.(11) we can say that there are ‘n’ number of energy levels and corresponding wave function. Thus we can say that each value of ‘E_n’ is known as “Eigen value” and the corresponding value of ‘Ψ_n’ is called Eigen function.

Energy levels of an Electron

For various values of ‘n’ we get various energy values of the electron. The lowest energy value (or) ground state energy value can be got by substituting n = 1 in Eqn.(11)

When n=1 ; we get; $E_1 = \frac{h^2}{8ma^2}$

n=2 ; we get; $E_2 = \frac{4h^2}{8ma^2} = 4E_1$

n=3 ; we get; $E_3 = \frac{9h^2}{8ma^2} = 9E_1$

n=4 ; we get; $E_3 = \frac{16h^2}{8ma^2} = 16E_1$

In general, $E_n = n^2 E_1 \dots\dots\dots (12)$

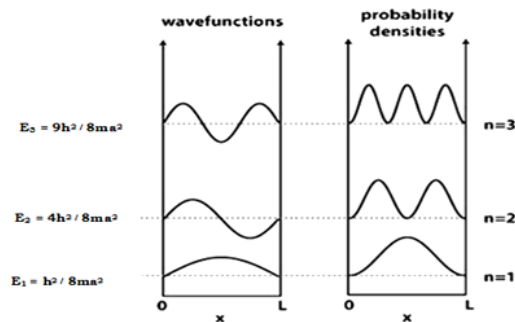


Fig.4.3. Energy levels of an electron

It is found from the energy levels $E_1, E_2, E_3, \dots$ the energy levels of an ' Ψ_n ' are discrete from the various energy levels mentioned above and it can be clearly said that quantum mechanics deals with discrete energy levels.

Normalization of the wave function:

It is the process by which the probability (P) of finding the particle (ie) inside the box can be done.

$$P = \int_0^a |\Psi|^2 dx = 1 \quad \dots\dots\dots (13)$$

For, a '1 – D' Box traveling in the 'x-direction' with the length 'a'.

Substitute Eqn. (8) in Eqn.(13) we get,

$$P = \int_0^a A^2 \sin^2 \frac{n\pi x}{a} dx = 1$$

$$A^2 \int_0^a \left| \frac{1 - \cos \frac{2n\pi x}{a}}{2} \right| dx = 1$$

$$A^2 \left[\frac{x}{2} - \frac{1}{2} \left[\frac{\sin \frac{2n\pi x}{a}}{\frac{2\pi n}{a}} \right] \right]_0^a = 1$$

[Since the second term of the integral becomes zero at both limits]

$$A^2 = \left[\frac{x}{2} \right]_0^a = 1, \text{ Thus, } \frac{A^2 a}{2} = 1$$

$$A^2 = \frac{2}{a} \quad (\text{or}) \quad A = \sqrt{\frac{2}{a}} \quad \dots\dots\dots (14)$$

Substituting the value of 'A' in Eqn. (8) we get the normalized wave function as

$$\Psi(n) = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a} \quad \dots\dots\dots (15)$$

4.9. PENETRATION OF A POTENTIAL BARRIER: TUNNEL EFFECT

Consider a potential barrier as shown in figure 4.4. In the region $x = 0$ to $x = a$ ($V > E$). According to classical mechanics, a steam of particles coming from the left would therefore be reflected. However, consider a potential barrier as shown in above figure 4.4.

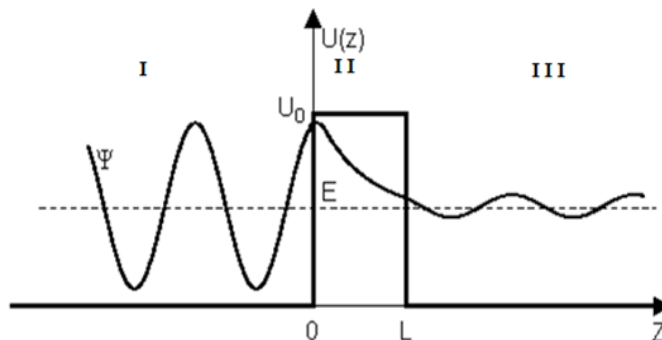


Fig.4.4. Penetration of a Potential barrier

In the region $x = 0$ to $x = a$ ($V > E$). According to classical mechanics, a stream of particles coming from the left would therefore be reflected. However, because of the wave nature of matter, there is some probability that the particle penetrates out of the other side of the barrier as we shall show, it can actually escape into a region $x > a$, where $E > V$. If the particle is impinging on the barrier with energy certainly less than the height of the potential barrier, it will not necessarily be reflected but there is always a probability that it may cross the barrier and continue its forward motion. This probability of crossing the barrier is called the **Tunnel effect**.

The Schrodinger's wave equation is

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \frac{2m(E-Vx)\Psi_1}{\hbar^2} = 0 \quad \dots\dots\dots (1)$$

The Schrodinger's wave equations for different region are: For the region I substitute $V(x) = 0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \frac{2mE\Psi_1}{\hbar^2} = 0 \quad \dots\dots\dots (2)$$

$$\frac{\partial^2 \Psi_1(x)}{\partial x^2} + \Psi_1 k^2 = 0 \quad \dots\dots\dots (3)$$

$$k^2 = \frac{2mE}{\hbar^2} \quad \dots\dots\dots (4)$$

The solution for Eqn. (3) is given by

$$\Psi_1 = Ae^{ikx} + Be^{-ikx} \quad \dots\dots\dots (5)$$

For the region II substitute $V(x) = V_0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} + \frac{2m(E-V_0)\Psi_2}{\hbar^2} = 0$$

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} - \frac{2m(V_0-E)\Psi_2}{\hbar^2} = 0 \quad \dots\dots\dots (6)$$

$$\frac{\partial^2 \Psi_2(x)}{\partial x^2} - \beta^2 \Psi_2 = 0 \quad \dots\dots\dots (7)$$

$$\beta^2 = \frac{2m(V_0-E)\Psi_2}{\hbar^2} \quad \dots\dots\dots (8)$$

The solution for Eqn. (7) is given by

$$\Psi_2 = Ce^{-\beta x} + De^{-\beta x} \quad \dots\dots\dots (9)$$

For region III substitute $V(x) = 0$ in Eqn. (1) we get,

$$\frac{\partial^2 \Psi_3(x)}{\partial x^2} + \frac{2mE\Psi_3}{\hbar^2} = 0 \quad \dots\dots\dots (10)$$

$$\frac{\partial^2 \Psi_3(x)}{\partial x^2} + k^2 \Psi_3 = 0 \quad \dots\dots\dots (11)$$

$$k^2 = \frac{2mE}{\hbar^2} \quad \dots\dots\dots (12)$$

The solution for Eqn.(12) is given below,

$$\Psi_3 = Ge^{ikx} + He^{-ikx} \quad \dots\dots\dots (13)$$

As there is no reflection in region III, the term He^{-ikx} is discarded.

$$\Psi_3 = Ge^{ikx} \quad \dots\dots\dots (14)$$

Here A, B, C, D and G are the constants whose values can be obtained by applying the following boundary conditions:

In order that Ψ and $\frac{\partial \Psi}{\partial x}$ be continuous at $x=0$, we have

$$(\Psi_1)_{x=0} = (\Psi_2)_{x=0} \quad \dots\dots\dots (15)$$

Applying the condition of Eqn.(15) and Eqn.(5) and Eqn.(9) and substituting $x=0$ we get.

$$A+B = C+D \quad \dots\dots\dots (16)$$

$$\left(\frac{\partial \Psi_1}{\partial x}\right)_{x=0} = \left(\frac{\partial \Psi_2}{\partial x}\right)_{x=\infty} \quad \dots\dots\dots (17)$$

Applying the condition of Eqn.(17) and equating Eqn.(5) and Eqn.(9) and substituting $x=0$ we get.

$$Ae^{ikx} + Be^{ikx} = Ce^{-\beta x} + De^{-\beta x}$$

$$\text{ie., } ikA - ikB = \beta C - \beta D \quad \dots\dots\dots (18)$$

$$ik(A-B) = \beta(C-D) \quad \dots\dots\dots (19)$$

$$(A-B) = \frac{\beta(C+D)}{ik} \quad \dots\dots\dots (20)$$

Adding Eqn. (16) from Eqn. (19) we get

$$2A = \left(1 + \frac{\beta}{ik}\right) C + \left(1 - \frac{\beta}{ik}\right) D \quad \dots\dots\dots (21)$$

Subtracting Eqn. (16) from (19) we get

$$2B = \left(1 - \frac{\beta}{ik}\right) C + \left(1 + \frac{\beta}{ik}\right) D \quad \dots\dots\dots (22)$$

Again in order that Ψ and $\partial \Psi / \partial x$ be continuous at $x=a$, we have,

$$(\Psi_2)_{x=a} = (\Psi_3)_{x=a} \quad \dots\dots\dots (23)$$

Applying the condition of Eqn. (22) and equating Eqn. (19) and Eqn.(14) and substituting $x = a$ we get

$$Ce^{-\beta x} + De^{-\beta x} = Ge^{ikx} \quad \dots\dots\dots (24)$$

$$\left(\frac{\partial \Psi_2}{\partial x}\right)_{x=a} = \left(\frac{\partial \Psi_3}{\partial x}\right)_{x=a} \quad \dots\dots\dots (25)$$

Applying the condition of Eq. (24) and equating Eq.(19) and Eq.(14) and substituting $x=a$ we get

$$\beta C e^{\beta a} - D \beta e^{-\beta a} = ik G e^{ika} \quad \dots\dots\dots (26)$$

$$\beta(C e^{\beta a} - D e^{-\beta a}) = ik G e^{ika}$$

$$(C e^{\beta a} - D e^{-\beta a}) = \frac{ik G e^{ika}}{\beta} \quad \dots\dots\dots (27)$$

Adding Eq.(24) and Eq.(25) we get,

$$C = \frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \quad \dots\dots\dots (28)$$

$$D = \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) e^{\beta a} \cdot Ge^{ika} \quad \dots\dots\dots (29)$$

To find the transmission Coefficient

$$2A = \frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \left(1 + \frac{ik}{\beta} \right) + \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) e^{\beta a} \cdot Ge^{ika} \left(1 - \frac{ik}{\beta} \right)$$

$$2A = \frac{Ge^{ika}}{2} \left[e^{-\beta a} \left[2 + \frac{\beta}{ik} + \frac{ik}{\beta} \right] + e^{\beta a} \left[2 - \frac{\beta}{ik} - \frac{ik}{\beta} \right] \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2e^{-\beta a} + \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] e^{-\beta a} + 2e^{\beta a} - \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] e^{\beta a} \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2(e^{\beta a} + e^{-\beta a}) + \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] (e^{-\beta a} - e^{\beta a}) \right]$$

$$A = \frac{Ge^{ika}}{4} \left[2(e^{\beta a} + e^{-\beta a}) - \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] (e^{\beta a} - e^{-\beta a}) \right]$$

$$A = Ge^{ika} \left[\frac{2(e^{\beta a} + e^{-\beta a})}{4} - \frac{1}{2} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \frac{(e^{\beta a} - e^{-\beta a})}{2} \right]$$

The formula for finding transmission coefficient $\frac{1}{T} = \left| \frac{A}{G} \right| \left| \frac{A}{G} \right|^*$

After simplification, the value of T is as follows

$$\frac{1}{T} = \left[1 + \left(\frac{V_0^2}{4E(V_0 - E)} \right) \sinh^2 \beta a \right] \text{ and } T = \left[1 + \left(\frac{V_0^2}{4E(V_0 - E)} \right) \sinh^2 \beta a \right]^{-1}$$

To find reflection coefficient

The formula for finding reflection coefficient $R = \left| \frac{B}{G} \right| \left| \frac{B}{G} \right|^*$

Where,

$$B = \frac{1}{2} \left(1 - \frac{ik}{\beta} \right) \left(\frac{1}{2} \left(1 + \frac{ik}{\beta} \right) e^{-\beta a} \cdot Ge^{ika} \right) + \left(1 + \frac{\beta}{ik} \right) \left(\frac{1}{2} \left(1 - \frac{\beta}{ik} \right) e^{\beta a} \cdot Ge^{ika} \right)$$

$$2B = \frac{Ge^{ika}}{2} e^{-\beta a} \left[\left[\frac{ik}{\beta} - \frac{\beta}{ik} \right] + \frac{1}{2} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} - \frac{ik}{\beta} \right] \right]$$

$$2B = \frac{i}{2} Ge^{ika} e^{-\beta a} \left[\left[\frac{ik}{\beta} + \frac{\beta}{ik} \right] + \frac{i}{2} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \right]$$

$$B = \frac{i}{4} Ge^{ika} e^{-\beta a} \left[\left[\frac{ik}{\beta} + \frac{\beta}{ik} \right] + \frac{i}{4} e^{\beta a} \cdot Ge^{ika} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right] \right]$$

$$B = \frac{i}{4} Ge^{ika} (e^{\beta a} \cdot e^{-\beta a}) \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right]$$

$$B = -\frac{i}{2} Ge^{ika} \frac{(e^{\beta a} - e^{-\beta a})}{2} \left[\frac{\beta}{ik} + \frac{ik}{\beta} \right]$$

Reflection coefficient, $R = \left| \frac{B}{G} \right| \left| \frac{B}{G} \right|^*$

After simplification, $R = \left[\frac{1}{4} \left(\frac{V_0^2}{E(V_0 - E)} \right) \sinh^2 \beta a \right]^{-1}$

4.10 THE TUNNEL DIODE

A p-n junction diode has an impurity concentration of about 1 part in 10^8 with this amount of doping, the width of the depletion layer, which constitutes a potential barrier at the junction, is of the order of 5 microns (5×10^{-4} cm). This potential barrier will not allow the flow of charge carriers from one side to the other.

The Tunneling Phenomenon:

Classically, a particle must have energy at least equal to the height of the barrier if it is to move from one side of the barrier to the other. However, if the width of the barrier is less than or equal to 10nm, the Schrodinger equation indicates that there is a large probability that an electron will penetrate through the barrier even though the energy E of the particle is less the height of the barrier V_0 . This quantum mechanical behavior is referred to as tunneling. The width of the barrier can be reduced to 10nm by high concentration of doping because the width of the junction barrier varies inversely as the square root of impurity concentration.

$$W \propto \frac{1}{\sqrt{N_A}}$$

Where, W – width of the barrier

N_A – Impurity concentration

In tunnel diode the concentration of impurity atoms is greatly increased, say to 1 part in 10^3 when compare to p–n diode so the width of the barrier in tunnel diode is reduced from 5 microns to less than 10nm or $100\text{\AA}$ (10^{-6} cm)

For very narrow barriers the wave function Ψ does not decay quick to zero before the other side is reached and Ψ stops decaying and becomes oscillatory again since $|\Psi|^2$ is larger than zero , there is finite probability that the electron will appear on the other side of the barrier as shown in below figure (4.5)

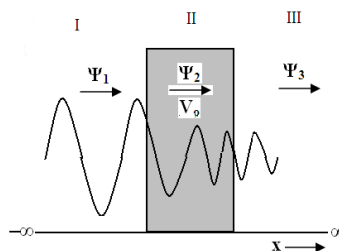


Fig. 4.5

Energy –Band structure of a highly doped p-n diode:

For tunneling to take place $E < V_0$ is a necessary but not a sufficient condition. It is also required that occupied energy states exist on the side from which the electron tunnels and that allowed empty states exist on the other side at the same energy level. Hence we must now consider the

energy –band picture when the impurity concentration is very high. The below figure 4.6 shows the energy band in a heavily doped p-n diode under open-circuited condition. From the figure 4.6, a diode which is doped heavily enough to make tunneling possible, the Fermi level E_F lies outside the forbidden band. Whereas for the lightly doped p-n diode the Fermi level E_F lies inside the forbidden energy gap.

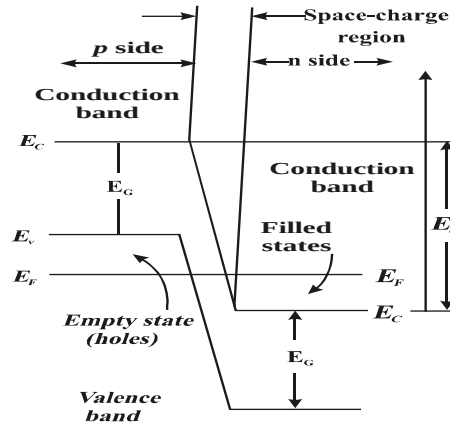
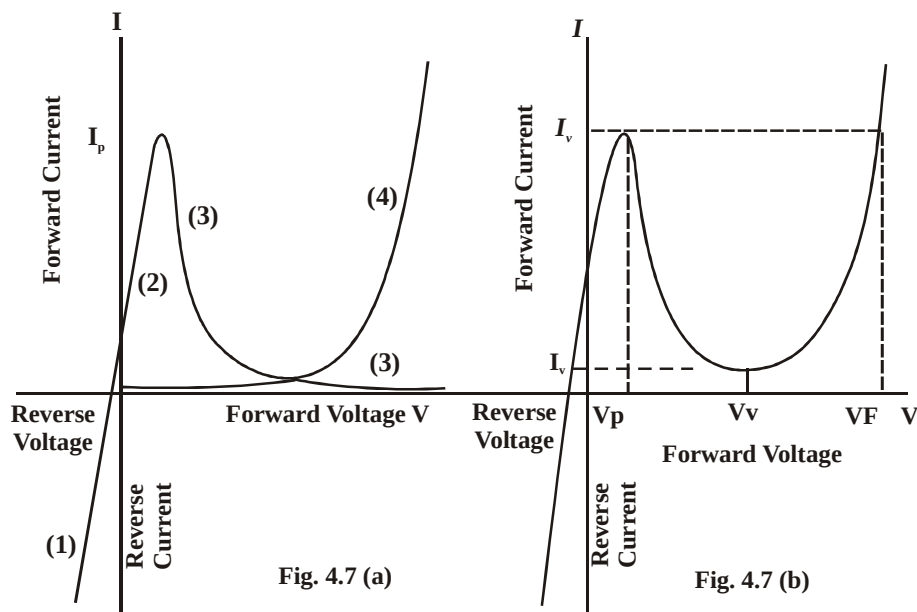


Fig. 4.6

In the figure the Fermi level E_F in the p-side is at the same energy as the Fermi level E_F in the n-side. Note that there are empty states on both side of the junction above E_F . Hence there can be no flow of charge in either direction across the junction, and the current is zero. A correct conclusion for an open circuit diode.

The volt –ampere characteristic of Tunnel Diode:

The following figure 4.7 (a) & (b) shows the volt –ampere characteristic of a tunnel diode under Reverse Bias and Forward Bias condition.



Reverse Bias Characteristic:

Let us consider that the p material is grounded and that a voltage applied across the diode, shifts the n-side with respect to the p-side. If a reverse-bias voltage is applied the n-side level must shift downward with respect to the p-side levels as indicated in the below figure 4.8.

We now observe that there are some occupied energy states [the dark shaded region] in figure 4.8 in the valence band of the p-side which lie at the same level as allowed empty states in the conduction band on the n-side. Hence these electrons will tunnel from the p to the n-side, giving rise to a reverse diode current. As the magnitude of the reverse bias increases, the heavily shaded dark area grows in size, causing the reverse current to increase as shown by number (2) of figure 4.7 (a).

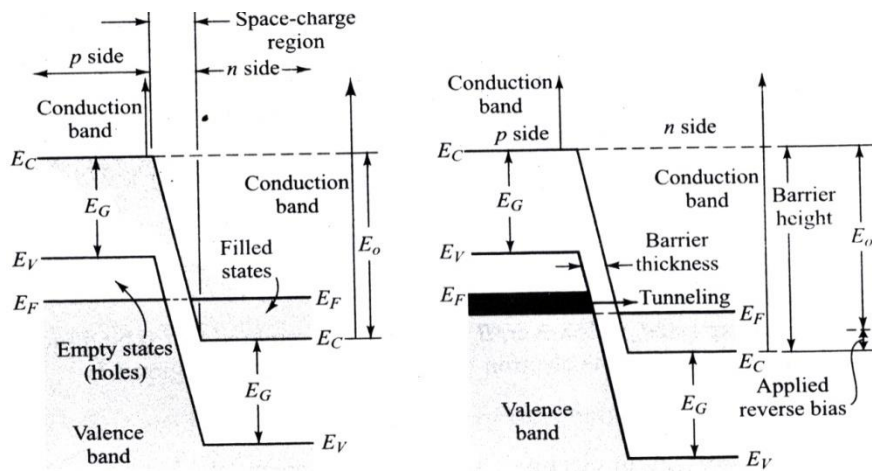
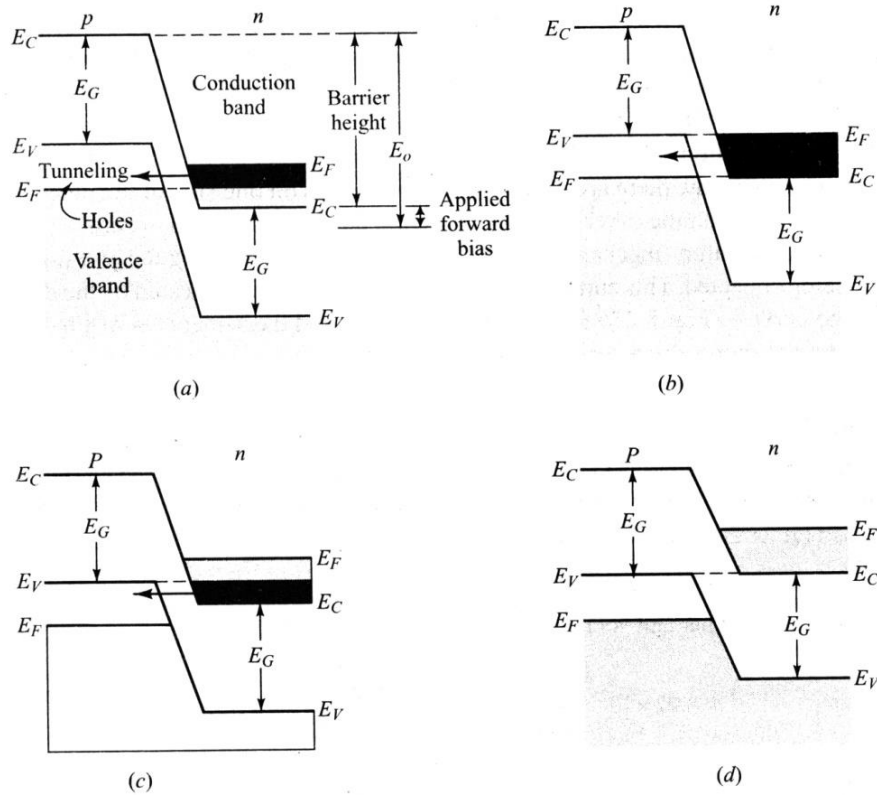


Fig. 4.8

Forward bias characteristics of tunnel diode

The following figures show the energy band structures of tunnel diode under forward bias nature. When forward bias voltage is applied to the diode the n-side levels must shift upward with respect to those on the p-side, and the energy-band picture for this situation is indicated in figure 4.9 (a). It is now evident that there are occupied states the conduction band of the n material (the heavily shaded dark levels) which are at the same energy allowed empty state (holes) in the valence band of a p-side. Hence electrons will tunnel from the n-side to the p material, giving rise to the forward current as indicated by the number 2 in figure 4.7 (a).

As the forward bias is increased further the condition shown in figure 4.9 (b) is reached. Now the maximum number of electrons can leave occupies states on the right side of the junction and turn through the barrier to empty states on the left side, giving rise to the peak current I_p and the corresponding voltage is called as peak voltage V_p as show in figure 4.7 (a). If still more forward bias is applied the situation in figure 4.9 (c) is obtained, and the tunnels current decreases, to a minimum called as valley current (I_v) and the corresponding voltage is called as valley voltage (V_v) which is shown as number 3 in figure 4.7 (a).



Finally, at an even larger forward bias, the band structure of figure 4.9 (d) is valid. Since now there are no empty allowed states on one side of the junction at the same energy as occupied states on the other side. The tunneling current must drop to zero.

In addition to the quantum-mechanical current described above, the regular p-n junction injection current is also being collected. This current is indicated by the dashed line number as 4 in figure 4.7(a). The figure 4.7 (b) is the sum of the solid and dashed curves of tunnel diode characters.

Pondicherry University 2 marks Question & Answer

1. What does mean by matter waves?(May 2014, May 2017)

During the movement of particles like electrons, protons, atoms and molecules, should have some type of wave motion associated with them. These waves are called as matter waves.

2. State the de Broglie hypothesis.(Jan 2013, May 2014)

According to de Broglie “the wave-particle dualism should not be restricted to radiation alone, but should also be extended to all the fundamental bodies of physics”

3. What do you mean de Broglie wavelength?(May 2013)

The wavelength of the wave associate with a moving particle is called as de Broglie wavelength. Particle with momentum ‘mv’ will have de Broglie wavelength of $\lambda = h / p$ (or) h / mv

4. What is de Broglie wavelength of an electron of an electron which has been accelerated from rest through a potential difference of 100 V.

(Jan 2011)

We have the relation $\lambda = 12.25 / \sqrt{V} \text{ \AA}$

In this case $V = 100$ volts. Hence $\lambda = 12.25 / \sqrt{100} = 1.225 \text{ \AA}$

5. State Heisenberg’s Uncertainty principle.

(Jan 2011, 2013, 2016, May 2013, 2017)

The principle states that the exact position and momentum of a particle cannot be determined simultaneously with a desired accuracy.

Taking Δx as the error in determining its position and Δp the error in determining its momentum at the same instant, then $\Delta x \cdot \Delta p = h / 2\pi$

6. Give the physical significance of ” ψ “. (Jan 2015, 2016)

Wave function “ ψ ” and its first order partial derivatives with respect to x, y, z must be continuous everywhere. “ ψ ” must be finite for all values of x, y and z . The wave function “ ψ ” beside being normalized must be single valued as $P(t)$ can have only one value at a particular place at a particular time.

7. Give the meaning of “ ψ ” in wave mechanics.(Jan 2014)

The wave function ψ is referred to probability amplitude for the position of the particle. While ψ itself has no physical interpretation the square of its absolute magnitude $|\psi|^2$ evaluate at a particular point at a particular time is proportional to the probability of experimentally finding there at that point.

8. What is the role of Schrodinger equation in wave mechanics?

Schrodinger equation, which is the fundamental equation of quantum mechanics in the same sense that the second law of motion is the fundamental equation of Newtonian mechanics, is a wave equation in the variable “ ψ ”. We can calculate the motion of a quantum-mechanical wave just as from Newton’s Law we can calculate the motion of a classical particle.

9. What do you mean by potential barrier?

A potential barrier corresponds to the force field acting on a particle being zero everywhere except in a limited region.

For $x < 0$, potential $V(x) = 0$.

For $x > 0$, potential $V(x) = V_0$

where V_0 is the height of the potential barrier.

10. Define the potential function of a potential step.

The potential function of a potential step or potential barrier as $V(x) = \{ 0 \text{ for } x < 0 \text{ and } V_0 \text{ for } x > 0 \}$ where V_0 is the height of the potential barrier.

11. State Tunnel effect.(Jan 2014, 2016)

If a particle is impinging on a barrier with energy less than the height of the potential barrier; it will not necessarily be totally reflected by the barrier but there is always a probability that it may cross the barrier and continue its forward motion.

12. Write a short note on properties of matter waves. (Jan 2016. May 2017)

- Lighter is the particle, greater is the wavelength associated with it.
- Smaller is the velocity of the particle, greater is the wavelength associated with it.
- If $v = 0$, then $\lambda = \text{infinity}$, i.e., wave become indeterminate and if $v = \text{infinity}$ then $\lambda = 0$. This indicates that the De-Broglie waves are generated by the motion of particle.
- These waves are produced in the particles whether the particles are charged particles or uncharged particles (in the equation $\lambda = h/mv$, there is no term representing the charge. It is independent of charge). This shows that these waves are not EM waves but are a new kind of waves.
- The velocity of matter wave is not constant since it depends on the velocity of the material particle whereas the velocity of EM wave is constant.
- The velocity of these waves is greater than velocity of light. They can travel faster than light.

13. Calculate the de-Broglie wavelength associated with a proton moving with a velocity of $1/10^{\text{th}}$ of velocity of light. Given the mass of proton is $1.67 \times 10^{-27} \text{ kg}$.(Jan 2016)

Solution

Mass of the proton $m = 1.67 \times 10^{-27} \text{ kg}$

Velocity of the proton $v = 1/10 \times \text{velocity of light}$

$$= 1/20 \times 3 \times 10^8$$

Velocity of light $v = 3 \times 10^8 \text{ m/s}$

De-Broglie wavelength $\lambda = h/mv$

$$\lambda = 6.626 \times 10^{-34} / 1.675 \times 10^{-27} \times 3 \times 10^7 = 1.3186 \times 10^{-14} \text{m}$$

De-Broglie wavelength $\lambda = 1.3186 \times 10^{-14} \text{m}$

ADDITIONAL PART – A Question & Answer

1. Explain the term degenerate energy level.

A set of two or more state having the same energy level is known as degenerate energy level.

2. Calculate the De-Broglie wavelength of an electron accelerated by a potential difference of 150 volt.

Given $h = 6.626 \times 10^{-34} \text{Js}$, $m = 9.1 \times 10^{-31} \text{kg}$

Charge of electron = $1.6 \times 10^{-19} \text{C}$

Solution:

$$\begin{aligned} \lambda &= h / \sqrt{2MeV} \\ &= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 1.6 \times 10^{-19} \times 9.1 \times 10^{-31} \times 150}} \\ \lambda &= 1.227 \times 10^{-10} \text{m}. \end{aligned}$$

3. Determine the velocity of electron having energy of 0.025 Mev.

Solution: we know that the kinetic energy of moving electron

$$\begin{aligned} \text{K.E} &= \frac{1}{2} mv^2 \\ v^2 &= \frac{2E}{m} \\ v &= \sqrt{\frac{2E}{m}} \\ v &= \sqrt{\frac{2 \times 0.025 \times 10^6}{9.1 \times 10^{-31}}} = 2.344 \times 10^{17} \text{ m/s} \end{aligned}$$

4. Calculate uncertainty of momentum and given uncertainty position of 0.1Å

$$\begin{aligned} \Delta x \cdot \Delta p_x &= h/2\pi = \hbar \\ \Delta p_x &= \hbar / \Delta x \\ &= 6.626 \times 10^{-34} / 0.1 \times 10^{-10} \\ \Delta p_x &= 6.626 \times 10^{-24} \text{ kg ms}^{-1} \end{aligned}$$

5. Mention the ideas which prompted de Broglie to propose his concept of Matter Waves.

i) Nature prefers symmetry. If radiation light can act like wave sometime and like a particle at other time, then material particles (eg: electrons, neutrons etc.) should also act as wave at sometime and particles in some other times.

ii) Our universe is fully composed of light and matter.

iii) Every moving particle is always associated with waves. The particle is controlled by this wave.

6. What are matter waves?

Every moving particle like electrons, protons, atoms and molecules has always associated with waves.

The particle is controlled by these waves and they are called Matter Waves.

PROBLEMS

1. An electron initially at rest, accelerated through a P.D of 5000 V Compare (i) The momentum and (ii) the de Broglie wavelength of the electron.

Solution

Let v be the velocity of the electron, then its K.E = $\frac{1}{2}mv^2 = Ve$

(i) Momentum of the electron

$$mv = \sqrt{2meV}$$

$$mv = \sqrt{2(9.11 \times 10^{-31}) 5000(1.6 \times 10^{-19})}$$

$$= 3.818 \times 10^{-23} \text{ kg ms}^{-1}$$

$$(ii) \frac{h}{mv} = \frac{6.625 \times 10^{-34}}{3.818 \times 10^{-23}} = 1.729 \times 10^{-11} \text{ m}$$

2. A microscope, using photons is employed to locate an electron in an atom to within a distance of 0.2 Å. What is the uncertainty in their momentum of the electron located in this way?

Solution

$$\Delta x = 0.2 \text{ Å} = 0.2 \times 10^{-10} \text{ m and } \Delta p = ?$$

$$\Delta p = \frac{h}{2\pi\Delta x}$$

$$\Delta p = \frac{6.625 \times 10^{-34}}{2\pi(0.2 \times 10^{-10})}$$

$$= 5.274 \times 10^{-24} \text{ kgms}^{-1}$$

3. An electron as a speed of 600ms⁻¹ with an accuracy of 0.005%. Compute the uncertainty with we can locate the position of the electron. $h = 6.6 \times 10^{-34} \text{ Js}$ and $m = 9.16 \times 10^{-31} \text{ kg}$. Momentum of the electron = $mv = 9.1 \times 10^{-31} \times 600 \text{ Kg. ms}^{-1}$.

$$\Delta p = \frac{0.005}{100} = (5 \times 10^{-5}) (9.1 \times 10^{-31} \times 600) \text{ kg.ms}^{-1}$$

Solution

from uncertainty principle, $\Delta x \Delta p = \frac{h}{2\pi}$

$$\begin{aligned}\Delta p &= \frac{h}{2\pi \Delta x} \\ &= \frac{6.6 \times 10^{-34}}{2\pi(5 \times 10^{-5})(9.1 \times 10^{-31} \times 600)} \\ &= 0.003846 \text{ m} = 3.84 \times 10^{-3} \text{ m}\end{aligned}$$

4. The life time of an excited state of an atom is about 10^{-8} sec. Calculate the minimum uncertainty in the determination of the energy of the excited state.

Solution

$$\begin{aligned}\text{We have, } \Delta E \cdot \Delta t &\geq \frac{h}{2\pi} \\ \Delta E &\geq \frac{h}{2\pi \Delta t} = \frac{6.6 \times 10^{-34}}{2\pi(10^{-8})} \\ \Delta E &\geq 1.0 \times 10^{-26} \text{ J} \\ &= 6.5 \times 10^{-8} \text{ eV}\end{aligned}$$

5. Calculate the permitted energy levels of an electron in a box of 1 width $1\text{\AA}$ (mass of electron $m = 9.16 \times 10^{-31} \text{ kg}$) (Jan 2016)

Solution

$$\text{permitted electron energies is given by } = \frac{n^2 h^2}{8ma^2}$$

for the first energy levels, $n = 1$ and hence we have,

$$\begin{aligned}E_1 &= \frac{1^2 (6.6 \times 10^{-34})^2}{8(9.1 \times 10^{-31} \times (10^{-10})^2)} \\ E_1 &= 38 \text{ eV.}\end{aligned}$$

This corresponds to the minimum energy

For $n=2$,

$$\begin{aligned}E_2 &= \frac{2^2 (6.6 \times 10^{-34})^2}{8(9.1 \times 10^{-31} \times (10^{-10})^2)} \\ E_2 &= 152 \text{ eV, } E_2 = 4 E_1\end{aligned}$$

Similarly,

$$E_3 = 342 \text{ eV} = 9 E_1.$$

6. An electron is bound in an one dimensional potential well of width $1\text{\AA}$, but of infinite wall height. Find its energy values in the ground state, and also in the first two states.

Width of the potential well $L = 1 \times 10^{-10} \text{ m}$; Find $E_0 = ?$, $E_1 = ?$ and $E_2 = ?$

Solution

The energy of the electron in one dimensional potential well is given by

$$E_n = \frac{n^2 h^2}{8ma^2}$$

Where, the values of E we get for n=1, 2 and 3 correspond respectively to E_0 , E_1 , and E_2

$$E_1 = \frac{1^2 h^2}{8ma^2}$$

$$E_1 = \frac{6.625 \times 10^{-34}}{8(9.1 \times 10^{-31})(10^{-10})^2} \text{ eV}$$

$$E_1 = 6.0314 \times 10^{-18} \text{ J} = 34.64 \text{ eV}$$

$$E_2 = \frac{2^2 h^2}{8ma^2} = 4 \frac{h^2}{8ma^2}$$

$$E_2 = 4 E_0$$

$$E_2 = 4 \times 6.0314 \times 10^{-18} \text{ J} = 150.54 \text{ eV}$$

$$E_3 = \frac{3^2 h^2}{8ma^2} = 9 \frac{h^2}{8ma^2} = 3 E_0$$

$$= 9 \times 6.0314 \times 10^{-18} \text{ J}$$

$$= 5.4282 \times 10^{-17} \text{ J}$$

$$E_3 = 338 \text{ eV}$$

PONDICHERRY UNIVERSITY PART B QUESTIONS

- 1. What are matter waves? Calculate the De- Broglie wavelength of an electron accelerated through a potential difference of 150 volts.**

(Jan 2013)

Answer: Refer Page No: 2 &20

- 2. Discuss how the description of a particle in terms of a wave packet leads to Heisenberg's uncertainty principle.**

Answer: Refer Page No: 4 to 5

- 3. Explain the physical significance of wave function.**

Answer: Refer Page No: 5 to 6

- 4. Derive the Schrodinger time dependent wave equation and hence obtain the time independent wave equation.**

(Jan 2011, Jan 2013, May 2013, May 2012, May 2017)

Answer: Refer Page No: 6 to 9

- 5. Explain the application of Schrodinger equation using one dimensional potential problem. Solve it to obtain eigen functions and show that eigen values are discrete. (May 2012, Jan 2014, 2015, 2016, May 2017)**

Answer : Refer Page No:9 to 12

- 6. Explain in detail the basic principle, construction and operation of a Tunnel Diode.(May 2014, Jan 2015, May 2017)**

Answer: Refer Page No: 16 to 19

- 7. Draw I-V characteristics of a tunnel diode under forward bias and reverse bias and explain it with the help of energy band diagram.(Jan 2016).**

Answer: Refer Page No: 16 to 19